

SYMPLIFY HEALTHCARE PLATFORM

Design System Consistency Audit

Version 1.7.4 • Comprehensive UI/UX Analysis

Prepared for: Symplify Engineering & Design Teams
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1. Executive Summary

This audit provides a comprehensive analysis of design system consistency across the Symplify Healthcare Operations Platform (v1.7.4). The platform spans 200+ page components across 15 major module directories, built on React with TypeScript, and uses a dual UI framework approach combining Bootstrap 5 and Ant Design.

The audit identified 47 distinct inconsistency patterns across 7 major categories. These range from critical accessibility violations that impact clinical workflows to minor cosmetic variations. The findings are organized by severity to enable a prioritized remediation plan.

8	14	16	9
Critical Issues	High Issues	Medium Issues	Low Issues

Key areas of concern include a fragmented color system with hardcoded values competing with CSS custom properties, inconsistent button sizing and spacing across modules, mixed icon library usage, and a dual-framework conflict between Bootstrap and Ant Design components that creates unpredictable visual behavior, particularly on clinical-facing pages.

2. Color System Analysis

2.1 Defined Palette vs. Actual Usage

The `_variables.scss` file defines a comprehensive token system with 90+ CSS custom properties. However, actual component usage frequently deviates from this system.

2.1.1 Hardcoded Color Values (Critical)

Multiple AI feature stylesheets bypass the design token system entirely, using hardcoded hex values that do not match any defined variable:

File / Component	Hardcoded Value	Expected Token	Severity
<code>_drug-interaction.scss</code>	#0d6efd (primary-color)	<code>var(--primary) = #2E37A4</code>	CRITICAL
<code>_drug-interaction.scss</code>	#28a745 (success green)	<code>var(--success) = #27AE60</code>	CRITICAL
<code>_drug-interaction.scss</code>	#dc3545 (danger red)	<code>var(--danger) = #EF1E1E</code>	CRITICAL
<code>_drug-interaction.scss</code>	#ffc107 (warning yellow)	<code>var(--warning) = #C9A227</code>	CRITICAL
<code>_voice-documentation.scss</code>	#dc3545, #6f42c1	<code>var(--danger)</code> , <code>var(--purple)</code>	HIGH
<code>_shift-handoff.scss</code>	#0d6efd, #667eea, #764ba2	No token equivalent	HIGH
<code>_ai-assistant.scss</code>	#7B1FA2 (Material purple)	No token equivalent	MEDIUM
<code>EmailPriorityBadge.tsx</code>	Inline: #DC2626, #F97316, etc.	<code>var(--clinical-*)</code> tokens	HIGH
<code>TriagePriorityBadge.tsx</code>	Inline: #F44336, #FF9800, etc.	<code>var(--clinical-*)</code> tokens	HIGH

Impact: The drug-interaction and voice-documentation features use Bootstrap 5's default palette (#0d6efd, #28a745, #dc3545) which conflicts with Symplify's custom brand palette (#2E37A4, #27AE60, #EF1E1E). This creates a visible "two-system" appearance when users navigate between core platform pages and AI features. Dark mode is also broken in these components since they don't reference CSS custom properties.

2.1.2 Duplicate Semantic Color Systems

The platform defines two overlapping semantic color systems:

System	Success	Danger/ Critical	Warning	Info
Platform Tokens (--success, etc.)	#27AE60	#EF1E1E	#C9A227	#2F80ED
Clinical Tokens (--clinical-*)	#16A34A	#DC2626	#CA8A04	#2563EB
Bootstrap Defaults (in AI SCSS)	#28a745	#dc3545	#ffc107	#0d6efd

Recommendation: Consolidate to a single semantic color system. The clinical tokens (--clinical-*) are the best candidate as they follow Tailwind CSS conventions and have proper light/dark mode variants. Map all Bootstrap defaults to these tokens. Remove hardcoded hex values from AI feature stylesheets.

3. Typography Inconsistencies

3.1 Font Size Fragmentation

The codebase declares Inter as the base font (`$font-family-base: 'Inter', sans-serif`) at 0.875rem (14px). However, multiple competing font-size patterns exist:

Pattern	Location	Values Found	Severity
Utility classes in <code>_ai-pages.scss</code>	Global scope, !important	fs-11 (11px), fs-12 (12px), fs-14 (14px), fs-15 (15px), fs-18 (18px), fs-22 (22px)	HIGH
Button <code>.btn</code> base font-size	<code>_buttons.scss</code>	0.75rem (12px) — conflicts with Bootstrap default	MEDIUM
Button <code>.btn-sm</code> dual declaration	<code>_buttons.scss</code>	font-size: 0.75rem AND font-size: 13px (same rule block)	MEDIUM
<code>.form-control</code> desktop vs mobile	<code>_forms.scss</code>	0.875rem desktop / 16px mobile (correct for iOS zoom)	LOW
<code>.form-select</code> font-size	<code>_forms.scss</code>	14px (hardcoded, not using rem)	LOW
Card title vs header-title	<code>_card.scss</code>	<code>.card-title</code> : 16px / <code>.header-title</code> : 1rem (same value, different units)	LOW
AI component font sizes	AI SCSS files	10px-18px range with px units, not rem	HIGH

3.2 Font Weight Inconsistencies

Variables define four weights (400, 500, 600, 700) but component usage mixes SCSS variable references with Bootstrap utility classes:

Component	SCSS/Variable Usage	Inline/Utility Usage	Severity
Page headers	<code>\$font-weight-semibold</code> (600)	<code>fw-bold</code> (700) in <code>PageHeader.tsx</code>	MEDIUM
Doctor details back link	<code>fw-semibold fs-14</code>	—	LOW
Create patient back link	<code>fw-bold</code>	Different weight than doctor details	MEDIUM
Table headers	<code>\$font-weight-semibold</code>	Consistent (good)	LOW
Badges	<code>\$font-weight-bold</code> (700)	Consistent (good)	LOW

Key finding: The PageHeader component uses h4.fw-bold while individual pages use h6.fw-semibold and h6.fw-bold inconsistently for the same back-navigation pattern. The create-patient page and doctor-details page are directly comparable workflows yet use different heading weights and sizes.

4. Button & Interactive Element Audit

4.1 Button Sizing Conflicts

Three competing button size definitions exist:

Source	Base .btn	.btn-sm	.btn-lg	.btn-icon	Severity
_buttons.scss	6px 10px / 0.75rem	6px 10px / 13px	8px 12px / 14px	32×32px	HIGH
_ai-pages.scss (override)	—	—	—	30×30px	HIGH
_ai-pages.scss .btn-xs	3px 10px / 11px	—	—	—	MEDIUM
Mobile (@media <md)	6px 12px / 0.8125rem / min-h:32px	min-h: 32px	—	min-h: 32px	MEDIUM

Issue: The _ai-pages.scss file redefines .btn-icon with a 30×30px size, conflicting with the 32×32px definition in _buttons.scss. Because _ai-pages.scss loads after _buttons.scss in the main.scss import order, the AI override wins globally, shrinking all icon buttons sitewide by 2px and falling below the WCAG minimum 32px touch target on desktop.

4.2 Touch Target Compliance

The mobile responsive rules enforce a 32px minimum touch target (WCAG 2.5.8 at Level AAA requires 44px). Analysis of touch targets:

Element	Desktop Size	Mobile Size	WCAG 2.5.5 (44px)	Severity
Table action items (.action-item > a)	22×22px	32×32px	Fails on all viewports	CRITICAL
.btn-icon base	32×32px (.btn) / 30×30px (AI)	32×32px	Fails AAA on all viewports	HIGH
.btn-icon.btn-xs	20×20px	No mobile override	Fails on all viewports	CRITICAL
.header-link	32×32px	No increase	Fails AAA on mobile	HIGH
Form checkboxes (mobile)	Default	20×20px	Fails AAA	MEDIUM

Clinical context: Healthcare workers frequently use touch devices while wearing gloves, which significantly reduces touch precision. The 22px action items in data tables are a critical usability issue for bedside documentation workflows.

5. Component Pattern Inconsistencies

5.1 Modal Border-Radius Conflict

The `_modal.scss` file declares `border-radius` twice on `.modal-content`:

Line 1: **`border-radius: 10px;`** (hardcoded)

Line 2: **`border-radius: $border-radius;`** (= 0.3rem ≈ 4.8px)

The second declaration wins, so modals render at ~5px radius while the intended design appears to be 10px. Mobile modals override to 12px. This creates three different modal corner treatments across viewports.

5.2 Card Component Inconsistencies

Property	Value in <code>_card.scss</code>	Value in AI Components	Severity
border-radius	5px	8px-16px (varies by AI component)	HIGH
padding (<code>.card-body</code>)	20px desktop / 16px-12px mobile	1.5rem (24px) in drug-interaction	MEDIUM
box-shadow	None (border only)	0 2px 12px rgba(0,0,0,0.08) in shift-handoff	MEDIUM
card-header background	transparent	linear-gradient in shift-handoff-widget	MEDIUM

5.3 Navigation & Back-Link Patterns

Two distinct back-navigation patterns exist across pages:

Page	Pattern Used	Heading Tag	Weight	Severity
Doctor Details	h6.fw-semibold.fs-14 with icon	h6	600	MEDIUM
Create Patient	h6.fw-bold with icon + "Patients"	h6	700	MEDIUM
Dashboard (PageHeader)	h4.fw-bold with breadcrumb	h4	700	LOW
Appointments (PageHeader)	h4.fw-bold with breadcrumb	h4	700	LOW

Recommendation: Standardize all detail/create pages to use the PageHeader component with consistent back-navigation. This also benefits accessibility by providing a uniform landmark structure.

5.4 Badge & Status Indicator Fragmentation

Four different badge/status systems are in use simultaneously:

System	Used In	Color Pattern	Severity
Bootstrap badge-soft-* utilities	Core platform pages	var(--state-transparent) bg + var(--state) text	LOW
TriagePriorityBadge (inline styles)	Appointments, AI pages	Hardcoded Material colors (bg-danger, etc.)	HIGH
EmailPriorityBadge (inline styles)	Email AI features	Tailwind palette (#DC2626, #22C55E)	HIGH
SeverityBadge (.severity-badge class)	Drug interaction checker	Bootstrap defaults (#d4edda, #28a745)	HIGH
.cc-severity-badge (AI assistant)	Chat assistant cards	Yet another color set	MEDIUM

Impact: A clinician viewing a “Critical” alert will see different shades of red depending on whether they are viewing a triage badge, email priority, drug interaction severity, or clinical alert. This undermines the semantic consistency that healthcare UIs critically depend on for patient safety.

6. Icon System Analysis

6.1 Multiple Icon Libraries

The platform loads 7 different icon font files:

Library	Font Files	Primary Usage	Severity
Tabler Icons (ti-*)	Via CDN/import	Primary UI icons throughout platform	LOW
Feather Icons	Feather.svg/ttf/woff	Older components, some navigation items	MEDIUM
Font Awesome	fontawesome-webfont.*	Sparse usage, legacy	MEDIUM
Material Icons	MaterialIcons-Regular.*	Very sparse usage	MEDIUM
Iconsax	iconsax.*	Sparse usage in specific modules	MEDIUM
Line Awesome	la-*.woff2	Minimal usage	MEDIUM
Bootstrap Icons	Via CSS (bootstrap-related)	Settings/theme UI	LOW

Performance impact: Loading 7 icon font families adds approximately 300–500KB of unnecessary font data. Tabler Icons is clearly the primary library (used in navigation, breadcrumbs, table sorting, and all AI components). The other 6 libraries should be audited for usage count and eliminated.

7. Spacing & Layout Inconsistencies

7.1 Spacing Token Adoption

The `_variables.scss` file defines a clean 4px-grid spacing system (`--space-1` through `--space-16`), but actual component usage overwhelmingly ignores these tokens:

Component	Defined Spacing	Actual Usage	Severity
Card body padding	<code>--spacing-card: 20px</code>	Hardcoded 20px in <code>_card.scss</code>	MEDIUM
Card header padding	<code>--spacing-card-header</code>	Hardcoded 0.9375rem / 1.25rem	MEDIUM
Modal header/footer	—	16px 20px (hardcoded)	LOW
Table cell padding	—	8px 16px (th) / 12px 16px (td) hardcoded	LOW
AI assistant chat	—	10px—14px (px throughout)	MEDIUM
Drug interaction checker	—	1.5rem / 1rem / 0.75rem (rem mix)	MEDIUM

Key finding: The spacing tokens exist but are essentially unused. The only consumer is the `.form-wizard-header` class. All other components use hardcoded pixel or rem values, defeating the purpose of having responsive spacing tokens that adjust across breakpoints.

7.2 Responsive Breakpoint Strategy Conflict

The `_mixins.scss` file documents a conflict: the platform uses both mobile-first (`respond-above`) and desktop-first (`respond-below`) patterns. The mixin file's own comment says “for new components, prefer `respond-above` (mobile-first) pattern” but approximately 80% of existing styles use `respond-below` (desktop-first).

The AI feature stylesheets (shift-handoff, AI assistant) use raw `@media` queries with hardcoded pixel breakpoints (480px, 768px, 1024px, 1440px) that don't align with the platform's defined breakpoint map (576px, 768px, 992px, 1200px, 1400px).

8. Framework Conflict: Bootstrap 5 vs. Ant Design

The platform uses both Bootstrap 5 (via SCSS and utility classes) and Ant Design (imported as React components). This dual-framework approach creates several conflicts:

Conflict Area	Bootstrap Pattern	Ant Design Pattern	Severity
Form selects	.form-select with custom styling	<Select> from antd with its own styles	HIGH
Date pickers	bootstrap-daterangepicker	<DatePicker> from antd	HIGH
Calendar	FullCalendar plugin	<Calendar> from antd on dashboard	MEDIUM
CSS variable naming	--bs-* (Bootstrap convention)	--ant-* (Ant convention)	MEDIUM
Modal styling	Bootstrap .modal with custom overrides	Ant Modal component (in some AI features)	HIGH
Rating component	None (custom stars in AI)	<Rate> from antd (smart-scheduler)	LOW
Spin/Loading	Custom CSS spinner in shift-handoff	<Spin> from antd in TriagePriorityBadge	MEDIUM

Root cause: The core platform was built as a Bootstrap 5 admin template (attributed to “Dreams Technologies” / ThemeForest in main.scss). The AI features and clinical enhancements were later added using Ant Design components, creating two visual systems. This is most apparent in the Appointments page where Bootstrap-styled form controls sit alongside the Ant Design DatePicker and Calendar.

9. Dark Mode Support Gaps

The `_variables.scss` defines comprehensive dark mode tokens under `:root[data-bs-theme=dark]`. However, multiple components will break in dark mode:

Component	Issue	Severity
Drug interaction checker	Uses hardcoded <code>#d4edda</code> , <code>#f8d7da</code> , <code>#fff3cd</code> — light-only colors	CRITICAL
Voice documentation	Uses hardcoded <code>#dc3545</code> on white assumed backgrounds	HIGH
Shift handoff summary	Uses <code>--card-bg</code> , <code>--bg-light</code> , <code>--text-primary</code> fallbacks that don't map to Symplify tokens	HIGH
AI assistant chat	Uses <code>var(--bs-body-bg)</code> and <code>var(--bs-body-color)</code> — Bootstrap vars, not Symplify vars	MEDIUM
Interaction alert backgrounds	<code>#d4edda</code> , <code>#fff3cd</code> , <code>#f8d7da</code> will appear as bright patches in dark mode	CRITICAL
Email priority badge	Inline styles with hardcoded <code>bgColor</code> values	HIGH
Clinical alert widget	<code>background-color: #fff</code> hardcoded on <code>.alert-item</code>	HIGH

10. Issue Severity Distribution

Category	Critical	High	Medium	Low	Total
Color System	4	4	1	0	9
Typography	0	2	3	3	8
Buttons & Touch Targets	2	2	2	0	6
Component Patterns	0	3	4	2	9
Icon System	0	0	5	2	7
Spacing & Layout	0	0	4	2	6
Dark Mode	2	3	1	0	6
TOTAL	8	14	16	9	47

11. Design System Recommendations

11.1 Unified Color Palette

Consolidate all three color systems (platform tokens, clinical tokens, Bootstrap defaults) into a single hierarchy:

Token Tier	Purpose	Naming Convention	Example
Primitive	Raw color values (never used directly)	--color-blue-600	#2E37A4
Semantic	Meaning-based tokens	--color-primary, --color-danger	Maps to primitives
Clinical	Healthcare-specific semantics	--clinical-critical, --clinical-stable	Extends semantic tier
Component	Scoped to specific UI patterns	--btn-primary-bg, --alert-success-bg	References semantic tokens

11.2 Typography Scale

Replace all ad-hoc font sizes with a rational scale anchored to the 14px base:

Token	Size	Use Case	Weight
--text-xs	0.6875rem (11px)	Timestamps, badges, fine print	500-600
--text-sm	0.75rem (12px)	Table headers, secondary labels	500-600
--text-base	0.875rem (14px)	Body text, form inputs, buttons	400-500
--text-md	1rem (16px)	Card titles, emphasized text	600
--text-lg	1.125rem (18px)	Section headers	600-700
--text-xl	1.25rem (20px)	Page titles	700
--text-2xl	1.5rem (24px)	Dashboard stat values	700-800

11.3 Component Standardization

Define a single source of truth for each reusable pattern:

Component	Proposed Standard	border-radius	Priority
Buttons (all sizes)	Align to 8px/12px/16px pad, min 32px desktop / 44px mobile	6px	P0 (Sprint 1)

Cards	20px padding, 8px radius, var(--border-color) border	8px	P0 (Sprint 1)
Modals	Fix dual border-radius, 16px header/footer, 20px body	12px	P1 (Sprint 2)
Badge / Status indicators	Single ClinicalBadge component using clinical tokens	4px / pill	P0 (Sprint 1)
Form inputs	All use rem units, min-height 44px on mobile	6px	P1 (Sprint 2)
Table action items	Minimum 32px desktop, 44px mobile	6px	P0 (Sprint 1)

11.4 Implementation Roadmap

Phase	Sprint(s)	Focus	Impact
Phase 0: Critical Safety	Sprint 1	Fix touch targets below 32px, unify clinical severity colors	8 Critical + 4 High issues
Phase 1: Token Migration	Sprint 2-3	Replace all hardcoded colors in AI SCSS with design tokens, fix dark mode	10 High + 6 Medium issues
Phase 2: Component Library	Sprint 4-5	Create shared ClinicalBadge, StandardButton, FormInput components	12 Medium + 4 Low issues
Phase 3: Icon Consolidation	Sprint 6	Audit & remove unused icon fonts, standardize on Tabler	5 Medium + 2 Low
Phase 4: Framework Resolution	Sprint 7-8	Migrate Ant Design components to Symplify-styled wrappers or replace	Long-term architecture

11.5 Governance Recommendations

1. **Lint rules:** Add Stylelint rules to flag hardcoded color hex values and px font sizes in SCSS files.
2. **Component library:** Establish a Storybook instance documenting every approved component variant with its props and visual states.
3. **PR review checklist:** Require design token usage verification in code review for any UI-touching changes.
4. **Accessibility CI:** Integrate axe-core or Lighthouse accessibility audits into the CI pipeline to catch touch-target violations automatically.
5. **Dark mode testing:** Add dark mode screenshots to visual regression tests, especially for all AI feature components.

End of Audit Report
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